

Personalized Briefing: Semantic Knowledge Graphs – Intersection of Dinis Cruz & Kerstin Clessienne’s Work

Executive Summary

Both **Kerstin Clessienne** and **Dinis Cruz** are thought leaders who recognize the transformative power of **semantic knowledge graphs** in unlocking the full potential of AI and data-driven decision making. Kerstin, an expert in marketing technology and strategy, has been advocating for knowledge graph-driven approaches to improve customer insight, personalization, and breaking down data silos in organizations. Dinis, a cybersecurity and GenAI expert, has been developing self-evolving semantic knowledge graph systems to capture complex knowledge and generate tailored content, even extending these graphs into **maps** (in the Wardley Mapping sense) to enhance strategic understanding. This briefing highlights Kerstin’s persona and graph-focused research, outlines Dinis’s work on semantic knowledge graphs (and his evolution toward Wardley Maps), and pinpoints where their research interests intersect. It also provides resources for Kerstin to explore Dinis’s work in more depth.

Kerstin Clessienne: Graph-Focused Persona & Key Contributions

Kerstin Clessienne is an internationally experienced advisor in applied marketing technology and organizational strategy, who explicitly brands herself as a *“Marketing Business Graph and Business AI Specialist.”* Her focus areas include **business modelling**, **marketing optimization**, customer experience (CX), and MarTech – all approached with a no-nonsense, human-centered philosophy ¹. In practice, Kerstin’s recent work has zeroed in on how **knowledge graphs and semantic AI** can elevate marketing and business intelligence:

- **Advocacy for Knowledge Graphs in AI:** Through her LinkedIn publications (e.g. *“Next Level GenAI”* series), Kerstin argues that integrating knowledge graphs with AI (specifically large language models) is a *“game-changer”* for creating truly personalized and context-aware experiences ². Unlike one-off prompt engineering or static fine-tuning approaches, knowledge graphs provide *structured, real-time adaptability* that enhances the efficiency and relevance of AI outputs ². She notes that KGs effectively give the AI a **“map” of meaning and context** to navigate user preferences, brand guidelines, and goals ³ – resulting in more on-point, creative, and non-generic responses. In her own words, *“Knowledge Graphs transform LLMs for creative and personalized work”*, enabling dynamic updates with new data or feedback without retraining ² ³. This perspective aligns with her marketing background: a well-structured ontology and graph of brand/audience knowledge can steer AI to generate content that resonates with each customer segment, at the right time with the right message.
- **Semantic Hubs and Breaking Data Silos:** Kerstin emphasizes that advanced GenAI initiatives in companies *require* a robust semantic data foundation – what she calls *“semantic lifting and a semantic hub.”* In her view, an organization’s internal knowledge (its *“company brain”*) needs to be organized in graphs/ontologies and connected to customer-facing systems in real-time ⁴. She cites research highlighting the pain of data fragmentation: *42% of companies struggle to*

quickly provide important data across departments, leading to inefficiency and slow decisions ⁵. Graph-based knowledge hubs address this by linking siloed data into a coherent whole. In fact, Gartner predicts that by 2025, **graph technologies** will be used in 80% of data analytics innovations, with **knowledge graphs** noted as a particularly relevant form ⁵. Companies that have adopted knowledge graphs already report significantly improved data quality and faster decision-making, validating her stance that KGs are becoming a “*central driver of innovation for modern organizations*” ⁶. Kerstin’s recent writings reinforce that organizations must invest in semantic knowledge infrastructure (ontologies, knowledge graphs, real-time data pipelines) to remain competitive. Marketing, she notes, can no longer operate in isolated MarTech platforms; it must reach **beyond**, integrating with enterprise knowledge graphs to ensure reliable, differentiated models and content ⁷. Without this, even the best AI efforts will be “*outdated and limited*” within a year ⁸.

- **Notable Works and Thought Leadership:** Kerstin co-authored the concept of “**Intelligente Transformation**” (Intelligent Transformation) and frequently shares insights on building “intelligent companies.” She combines strategic marketing know-how with deep appreciation for data semantics. For example, her *Semantic AI – Next Level GenAI* posts (Parts 1 and 2) detail the importance of semantic understanding in AI solutions. She stresses educating domain experts on data semantics and encourages cross-functional growth (“making people grow together”) to unleash creativity backed by solid data foundations ⁹ ⁴. Her thought leadership extends to explaining “*What exactly is a knowledge graph?*” to business audiences and highlighting use-cases of graphs in customer journey optimization, persona modeling, and content personalization. In summary, Kerstin’s persona is that of a **marketing technologist who bridges human-centric strategy with graph-powered AI**, always advocating for meaningful context over buzzwords. Her contributions set the stage for why semantic knowledge graphs are indispensable in modern marketing and decision-support – a viewpoint very much shared and complemented by Dinis’s work.

Dinis Cruz: Semantic Knowledge Graphs & Evolution into Wardley Maps

Dinis Cruz is a seasoned cybersecurity leader and GenAI expert (Founder of The Cyber Boardroom and Chief Scientist at Glasswall) known for pioneering the use of knowledge graphs and AI in security and organizational knowledge management ¹⁰. Over the past few years, his focus has expanded from traditional security/taxonomy challenges into building **self-evolving semantic knowledge graphs** and leveraging **Wardley Maps** for strategic insight. Key aspects of Dinis’s work include:

- **From Static Ontologies to Living Knowledge Graphs:** Dinis’s journey into semantic graphs began with frustration at the limitations of static, top-down ontologies and taxonomies. In practice, he found rigid taxonomies often “*don’t reflect the organic and evolutionary nature of reality*” and quickly break when confronted with real-world complexity ¹¹. As a vCISO, he experimented with graph visualization to map security knowledge, but maintaining these structures manually was onerous and hard to scale. His breakthrough insight (through developing projects like *The Cyber Boardroom*) was that the solution lies not in chasing perfect up-front ontologies, but in “**self-improving knowledge graphs.**” In his own words, the goal shifted to creating graphs that evolve and update organically with the input of both humans and AI, rather than enforcing a brittle structure ¹² ¹³. This approach uses GenAI to keep the knowledge graph in sync and enriched: AI can suggest new relationships, classify information on the fly, and even help reconcile different vocabularies used by different teams. Dinis emphasizes that the *synergy of GenAI with graphs* enables knowledge structures that continuously adapt and

“learn,” solving problems of scale and maintenance that once seemed intractable ¹³ . Crucially, this means the knowledge graph can serve as a dynamic “single source of truth” even in fast-changing domains.

- **“Graphs of Graphs” and Bridging Domain Ontologies:** A distinctive element of Dinis’s research is the concept of linking multiple graphs and ontologies together – essentially **graphs-of-graphs** or an “ontology of ontologies.” Organizations often have various overlapping data domains (each team or use-case may have its own mini-ontology). Instead of forcing everyone into one standard (which he notes is often impossible or short-lived), Dinis builds systems to let these sub-graphs co-exist and connect. GenAI acts as a mediator that can *“bridge between team ontologies without forcing standardisation,”* allowing different departments to share knowledge without losing their domain-specific context ¹⁴ . This respects diverse perspectives while still maintaining interconnections. The result is a federated knowledge network where, for example, marketing’s terminology and product development’s terminology can be mapped and translated by the AI-driven graph. Such an evolving semantic hub directly tackles the fragmentation issue that Kerstin highlighted: it reduces silos by linking data on-the-fly, rather than requiring a single monolithic schema. Dinis also implements **human-in-the-loop feedback loops**, where usage of the knowledge graph (e.g. generating a report or answering a query) provides feedback to improve the graph’s accuracy and relevance over time ¹⁵ . This continuous refinement process is akin to an organism learning – aligning with his philosophy of organic, evolving knowledge structures.
- **Generating Personalized Narratives and Content:** While Kerstin applies knowledge graphs to personalize marketing content, Dinis has been applying similar principles to generate tailored **briefings and narratives** for different stakeholders. He observed that merely having a knowledge graph isn’t enough – you need to turn that graph data into consumable insights for people (stories, reports, answers) in order to validate and refine the knowledge. One of his focal projects, **MyFeeds.ai**, is built on this idea. MyFeeds uses semantic knowledge graphs on the back-end to connect information across many dimensions (topics, authors, sources, timelines, etc.), and then GenAI uses those graphs to generate highly personalized content streams ¹⁶ . Essentially, MyFeeds can produce a custom “feed” or briefing for a user that is context-rich and sourced from the graph’s knowledge. According to Dinis, this progression *“from metadata to stories”* is deliberate – metadata gives basic facts, graphs give context, and finally **stories** (narratives) present the information in a human-friendly way ¹⁷ ¹⁶ . By combining a semantic graph with generative AI, MyFeeds can create outputs that have both the accuracy of structured data and the fluidity of natural language. This mirrors Kerstin’s aim of more intelligent content personalization, and demonstrates how knowledge graphs can power not just backend data integration but also the front-end creative process.
- **Wardley Mapping: Evolving Graphs into Strategic Maps:** A notable aspect of Dinis’s evolution is his incorporation of **Wardley Maps** (a strategic mapping technique) into his knowledge graph work. In Dinis’s view, a *“map”* is essentially a graph with an added dimension: spatial position to convey meaning (such as stage of evolution or value chain position) ¹⁸ . Whereas a plain graph just shows relationships, a map organizes those elements on a landscape, making patterns and gaps much more apparent. Dinis has argued that to truly gain **situational awareness** from a complex knowledge domain, one must *“evolve [graphs] into maps”* ¹⁹ . For example, in a Wardley Map of a technology strategy, the y-axis might represent the value chain and the x-axis shows evolution from novel idea to commodity. Each node (which could be something like a capability, component, or knowledge element) is placed in this space. By mapping his knowledge graphs onto such strategic frameworks, Dinis can visualize how certain knowledge or capabilities progress and where the gaps are. As he notes, *in concept maps like Wardley Maps, the position of*

nodes signifies their stage of evolution (from genesis to commodity) ²⁰ – something a basic graph cannot communicate. This spatial context is powerful: it uncovers trends and dependencies that remain hidden in a normal node-edge diagram, thus helping teams make better decisions ²¹ ²². Dinis's use of Wardley Maps complements his semantic graphs by providing an **additional layer of meaning (movement, position, evolution)** on top of the raw relationships. In practice, this means Dinis not only connects knowledge in a graph, but also plots it against strategic dimensions (technology maturity, business value, etc.), offering a holistic view. This evolution from graphs to maps is a unique aspect of his work that Kerstin may find intriguing, especially as it could apply to marketing strategy (e.g., mapping customer knowledge or capabilities along a journey of maturity). It represents Dinis's commitment to not just gather knowledge, but to **contextualize it visually for strategic insight**.

- **Notable Projects & Publications:** Dinis has documented much of his research and experimentation openly. He launched an open-access repository of his research papers and presentations at docs.diniscruz.ai, to collate his work on semantic graphs, AI, and security in one place ²³. Some of his recent publications (March 2025) include *"Beyond Static Ontologies: How GenAI Powers Self-Improving Knowledge Graphs"* ¹¹ ²⁴ – an in-depth look at using GenAI to maintain and enrich knowledge graphs – and *"How Metadata, Graphs, Maps, and Stories Build Provenance Ecosystems"* ¹⁸ ¹⁷, which outlines the progression from raw data to semantic graphs to mapping and narrative generation. Additionally, Dinis has shared practical demos, such as using his **Serverless MGraph-DB** (a graph database he published) to maintain multiple interconnected semantic graphs across different stakeholder needs ²⁵. In the OWASP security community, he showcased how graph-based knowledge management combined with GenAI can scale security expertise (*"Semantic OWASP" talk*). In summary, Dinis's body of work centers on building **intelligent knowledge ecosystems** – where graphs are not static diagrams, but living systems that collaborate with AI and humans to yield actionable insights, and maps are used to orient and guide strategy.

Intersection of Research Interests

Given the above, it's clear that **Kerstin Clessienne and Dinis Cruz share a vision** for leveraging semantic knowledge graphs to create smarter organizations and AI systems. Key intersections in their interests and focus include:

- **Semantic Context as Key to AI:** Both see **semantic enrichment** (ontologies, knowledge graphs) as critical for AI to move beyond shallow pattern-matching. Kerstin highlights that any advanced AI in business *"will not be truly feasible without semantic lifting and a semantic hub"* ⁴, and likewise Dinis builds systems that infuse AI with graph-linked context so it can produce relevant, non-generic output. In essence, both are solving the same problem of *imparting knowledge and context to AI*, whether it's to generate a marketing message or a board-level security report.
- **Breaking Down Data/Knowledge Silos:** Both address the challenge of fragmented knowledge in organizations. Kerstin points to the serious inefficiencies caused by data silos (with research data to back it) and advocates for unified knowledge graphs to connect departments ⁵ ⁶. Dinis's approach of linking multiple domain-specific graphs with GenAI mediation is a direct solution to this fragmentation ¹⁴. In different language, they are both working toward an **integrated "company brain."** The concept of a *central but evolving knowledge hub* is a shared theme – Kerstin calls it a semantic hub; Dinis calls it interconnected self-improving graphs –

ultimately aiming for the same outcome: quick access to the right knowledge across the enterprise.

- **Personalization and Dynamic Content:** A striking overlap is how both use knowledge graphs to drive **personalized content and insights**. Kerstin discusses using knowledge graphs to tailor content to the right person at the right time (e.g. personalized customer journeys and brand messaging) ² ³. Dinis uses them to tailor briefings and stories to different stakeholders (through MyFeeds.ai and similar tools) ¹⁶. In both cases, the graph serves as a memory of “who is who and what context matters,” enabling AI to generate output that is much more targeted than it would be with generic training data alone. This intersection suggests potential collaboration – for example, Dinis’s techniques for automated story generation could enhance marketing personalization, and Kerstin’s domain ontologies in marketing could inform how Dinis’s systems model personas and content.
- **Knowledge Graphs as Future-Proofing:** Both experts are essentially preparing organizations for the future where knowledge graphs play a central role. They cite and agree with industry predictions that knowledge graph technology will underpin the next wave of AI and data innovation ⁵ ⁶. Kerstin provides the strategic rationale for marketing and customer experience, while Dinis provides the technical architecture to implement it at scale. They both recognize that organizations embracing semantic graphs will gain in agility (real-time updates, continuous learning) and decision quality, whereas those sticking to rigid or siloed data models will fall behind ⁸ ¹³.
- **Wardley Maps and Strategic Alignment:** An area of possible cross-pollination is Dinis’s use of **Wardley Maps** in conjunction with knowledge graphs. While Kerstin’s publications have not explicitly mentioned Wardley Mapping, her focus on *strategy and intelligent transformation* indicates she appreciates strategic frameworks. Dinis’s work shows how mapping capabilities can complement graph data by adding a view of **evolution and importance**. This could intersect with Kerstin’s interest if, for example, mapping the maturity of marketing capabilities or customer needs could yield insights for marketing strategy. Both ultimately care about **making sense of complexity** – Kerstin through logical graph models and human-centric narratives, Dinis through graph-to-map transformations and storytelling. Together, their ideas suggest an end-to-end pipeline: collect and interlink knowledge (graph), visualize strategy and position (map), and communicate insights (story). It’s a cohesive vision of intelligent enterprise decision-making that both are contributing to from different angles.

In summary, **Kerstin and Dinis converge on the principle that knowledge – structured, connected, and contextual knowledge – is the cornerstone of effective AI and digital strategy**. Kerstin’s marketing and business perspective and Dinis’s technical and strategic perspective are highly complementary. Both are effectively working on building the “**semantic layer**” for AI (Kerstin calls it the semantic hub; Dinis implements it via evolving graphs and maps). They aim to empower organizations to leverage AI in a way that is *grounded in truth, context, and strategy*, rather than generic or siloed. This shared ground provides an excellent basis for dialogue and collaboration between them.

Resources for Further Exploration

To facilitate deeper learning and collaboration, here are some resources through which Kerstin can explore Dinis's work (with focus on the semantic graph and mapping themes):

- **Dinis's Research Repository:** *docs.diniscruz.ai* – An MKDocs-based public site where Dinis publishes his research documents, presentations, and articles. This includes writings on semantic knowledge graphs, Wardley mapping, and GenAI experiments (as Dinis noted, it's a growing collection of his work-in-progress materials) ²³. Browsing this site will give insight into his past projects (e.g. OWASP security graph initiatives) and evolving ideas.
- **"How It Works" Series on MyFeeds.ai:** Dinis's startup **MyFeeds.ai** has a blog/tag called "How it works" that dives into the architecture and concepts behind its semantic knowledge graph approach. In these posts, Dinis explains how human expertise and GenAI combine in MyFeeds to build *"rich, flexible knowledge structures that continuously improve."* This is a practical look at implementing a self-improving knowledge graph for content personalization ²⁶, directly relevant to Kerstin's interest in using graphs for better content and customer engagement.
- **Key Articles by Dinis:** Two highly relevant LinkedIn Pulse articles by Dinis are worth reading in full: **"Beyond Static Ontologies: How GenAI Powers Self-Improving Knowledge Graphs"** (Mar 31, 2025) and **"How Metadata, Graphs, Maps, and Stories Build Provenance Ecosystems in the GenAI Semantic Web."** These pieces lay out Dinis's philosophy and approach in a narrative form. For instance, the former details moving from fixed taxonomies to organic graphs with GenAI bridges ¹¹ ¹³, and the latter illustrates the data-to-graph-to-map progression with Wardley Maps examples ¹⁸ ¹⁷. They provide a conceptual framework that resonates with many of Kerstin's themes (e.g., the importance of context and provenance in AI).
- **Talks and Presentations:** Dinis has presented on related topics in the past – for example, at OWASP London (on scaling security knowledge with GenAI and graphs) ²⁷ and at Map Camp (on using Wardley Maps for security strategy). These presentations (some available on YouTube or Slideshare) show real-world applications of his ideas. While they are in the security domain, the underlying methods (using graphs to model knowledge, and maps to plan strategy) are transferable to marketing and business transformation contexts that Kerstin operates in.

By exploring these resources, Kerstin can get a comprehensive view of Dinis Cruz's work with **semantic knowledge graphs and Wardley Maps**. This will not only provide a solid introduction to Dinis's current and past research on graphs but also spark ideas on how their mutual interests might cross-fertilize. Given Kerstin's own extensive experience with knowledge graphs for marketing and AI, this shared understanding can serve as a foundation for a rich exchange of ideas or potential collaboration, bridging the gap between marketing intelligence and technical innovation in knowledge-centric AI.

¹ ⁴ ⁷ ⁸ ⁹ Next Level GenAI Part 1: How much do you need to understand as a domain... | Kerstin Clessienne

https://www.linkedin.com/posts/kerstinclessienne_next-level-genai-part-1-how-much-do-you-activity-7329173598617616384-z43B

² ³ #design #content #prompting #adaptability #efficiency #map #triples... | Kerstin Clessienne
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